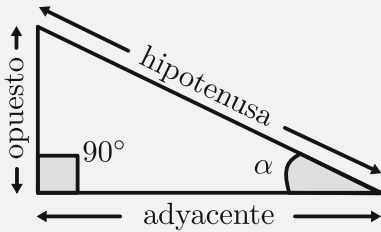


TRIGONOMETRÍA

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$$\text{S} \frac{o}{h} \quad \text{sen } \alpha = \frac{\text{opuesto}}{\text{hipotenusa}}$$

$$\text{C} \frac{a}{h} \quad \text{cos } \alpha = \frac{\text{adyacente}}{\text{hipotenusa}}$$

$$\text{T} \frac{o}{a} \quad \text{tan } \alpha = \frac{\text{opuesto}}{\text{adyacente}}$$

$\angle \alpha$ grados	π radianes	$\text{sen } \alpha$	$\text{cos } \alpha$	$\text{tan } \alpha$
0°	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	—
180°	π	0	-1	0
270°	$\frac{3\pi}{2}$	-1	0	—

IDENTIDADES TRIGONOMÉTRICAS

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Recíprocas

$$\operatorname{sen} \alpha = \frac{1}{\operatorname{csc} \alpha} \quad \operatorname{csc} \alpha = \frac{1}{\operatorname{sen} \alpha}$$

$$\cos \alpha = \frac{1}{\sec \alpha} \quad \sec \alpha = \frac{1}{\cos \alpha}$$

$$\tan \alpha = \frac{1}{\cot \alpha} \quad \cot \alpha = \frac{1}{\tan \alpha}$$

Pitagóricas

$$\operatorname{sen}^2 \alpha + \cos^2 \alpha = 1$$

$$1 + \tan^2 \alpha = \sec^2 \alpha$$

$$1 + \cot^2 \alpha = \operatorname{csc}^2 \alpha$$

Cociente

$$\tan \alpha = \frac{\operatorname{sen} \alpha}{\cos \alpha} \quad \cot \alpha = \frac{\cos \alpha}{\operatorname{sen} \alpha}$$

Ángulos opuestos

$$\operatorname{sen}(-\alpha) = -\operatorname{sen} \alpha$$

$$\cos(-\alpha) = \cos \alpha$$

$$\tan(-\alpha) = -\tan \alpha$$

Ángulos dobles

$$\operatorname{sen}(2\alpha) = 2 \operatorname{sen} \alpha \cos \alpha$$

$$\cos(2\alpha) = \cos^2 \alpha - \operatorname{sen}^2 \alpha$$

$$\tan(2\alpha) = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

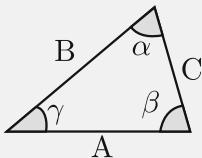
Suma y resta

$$\operatorname{sen}(\alpha \pm \beta) = \operatorname{sen} \alpha \cos \beta \pm \cos \alpha \operatorname{sen} \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \operatorname{sen} \alpha \operatorname{sen} \beta$$

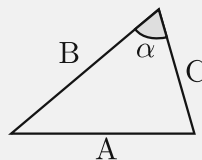
$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$$

Ley de senos



$$\frac{A}{\operatorname{sen} \alpha} = \frac{B}{\operatorname{sen} \beta} = \frac{C}{\operatorname{sen} \gamma}$$

Ley de cosenos



$$A^2 = B^2 + C^2 - 2 BC \cos \alpha$$

INTEGRALES

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$$\int dx = x + C$$

$$\int k dx = kx + C$$

$$\int (u \pm v) dx = \int u dx \pm \int v dx$$

$$\int k f(x) dx = k \int f(x) dx$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C; \quad n \neq -1$$

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int e^x dx = e^x + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C; \quad a > 0, a \neq 1$$

$$\int \ln x dx = x \ln x - x + C$$

$$\int \sen x dx = -\cos x + C$$

$$\int \cos x dx = \sen x + C$$

$$\int \tan x dx = -\ln |\cos x| + C$$

$$\int \cot x dx = \ln |\sen x| + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

$$\int \csc x dx = \ln |\csc x - \cot x| + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

$$\int \sen^2 x dx = \frac{1}{2}x - \frac{1}{4}\sen(2x) + C$$

$$\int \cos^2 x dx = \frac{1}{2}x + \frac{1}{4}\sen(2x) + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \csc^2 x dx = -\cot x + C$$

$$\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsen\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln |x + \sqrt{x^2 - a^2}| + C$$

$$\int \frac{x}{a^2 + x^2} dx = \frac{1}{2} \ln(a^2 + x^2) + C$$

$$\int u dv = uv - \int v du$$

L = Logarítmicas

I = Inversas

A = Algebraicas

T = Trigonómicas

E = Exponenciales

$$\int_a^b f(x) dx = F(b) - F(a)$$

DERIVADAS

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$$\frac{d}{dx}(k) = 0; \quad k \text{ es constante}$$

$$\frac{d}{dx}(x) = 1$$

$$\frac{d}{dx}(k u) = k u'$$

$$\frac{d}{dx}(x^n) = n x^{n-1}$$

$$\frac{d}{dx}(u^n) = n u^{n-1} \cdot u'$$

$$\frac{d}{dx}(\sqrt{u}) = \frac{u'}{2\sqrt{u}}$$

$$\frac{d}{dx}(\sqrt[n]{u}) = \frac{u'}{n \sqrt[n]{u^{n-1}}}$$

$$\frac{d}{dx}\left(\frac{1}{u}\right) = -\frac{u'}{u^2}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$$

$$\frac{d}{dx}(u \pm v) = u' \pm v'$$

$$\frac{d}{dx}(u v) = u'v + uv'$$

$$\frac{d}{dx}(e^u) = e^u \cdot u'$$

$$\frac{d}{dx}(a^u) = a^u \ln a \cdot u'$$

$$\frac{d}{dx}(u^v) = u^v \left(v' \ln u + v \frac{u'}{u} \right)$$

$$\frac{d}{dx}(\ln u) = \frac{u'}{u}$$

$$\frac{d}{dx}(\log_a u) = \frac{u'}{u \ln a}$$

$$\frac{d}{dx}(\sin u) = \cos u \cdot u'$$

$$\frac{d}{dx}(\cos u) = -\sin u \cdot u'$$

$$\frac{d}{dx}(\tan u) = \sec^2 u \cdot u'$$

$$\frac{d}{dx}(\cot u) = -\csc^2 u \cdot u'$$

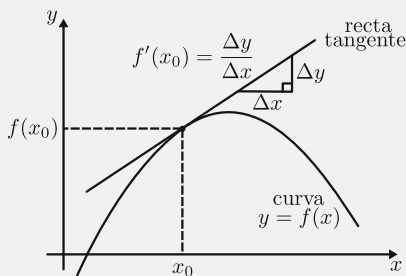
$$\frac{d}{dx}(\sec u) = \sec u \tan u \cdot u'$$

$$\frac{d}{dx}(\csc u) = -\csc u \cot u \cdot u'$$

$$\frac{d}{dx}(\arcsen u) = \frac{u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx}(\arccos u) = -\frac{u'}{\sqrt{1-u^2}}$$

$$\frac{d}{dx}(\arctan u) = \frac{u'}{1+u^2}$$



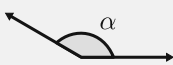
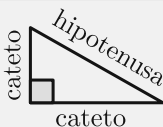
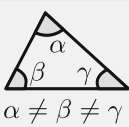
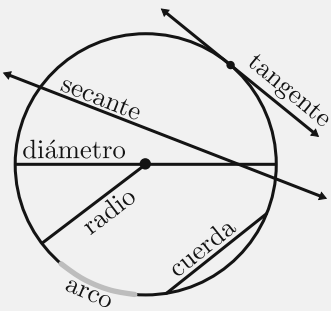
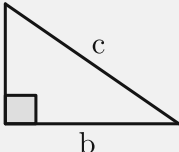
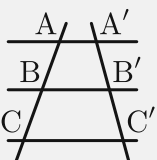
$$\frac{d}{dx}f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

GEOMETRÍA

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Agudo  $0 < \alpha < 90^\circ$	Recto  $\alpha = 90^\circ$	Obtuso  $90 < \alpha < 180^\circ$	Llano  $\alpha = 180^\circ$
Complementarios  $\alpha + \beta = 90^\circ$	Suplementarios  $\alpha + \beta = 180^\circ$	Paralelas 	Perpendiculares 
△ Equilátero 	△ Isósceles 	△ Rectángulo 	△ Escaleno  $\alpha \neq \beta \neq \gamma$
Líneas notables del triángulo		Partes del círculo	
Bisectriz  Incentro	Mediana  Baricentro		
Mediatriz  Circuncentro	Altura  Ortocentro		
Teorema de Pitágoras		Teorema de Tales	
 $a^2 + b^2 = c^2$		 $\frac{AB}{A'B'} = \frac{BC}{B'C'}$	

ÁLGEBRA

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$$a^0 = 1$$

$$a^{-n} = \frac{1}{a^n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

$$a^m \cdot a^n = a^{m+n}$$

$$\sqrt[n]{a} \sqrt[n]{a} = a^{\frac{1}{n} + \frac{1}{n}}$$

$$(ab)^n = a^n b^n$$

$$\sqrt[n]{ab} = \sqrt[n]{a} \sqrt[n]{b}$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$(a^m)^n = a^{mn}$$

$$\sqrt[n]{\sqrt[n]{a}} = \sqrt[n^2]{a}$$

$$\log_b(xy) = \log_b x + \log_b y$$

$$\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$$

$$\log_b(x^k) = k \log_b x$$

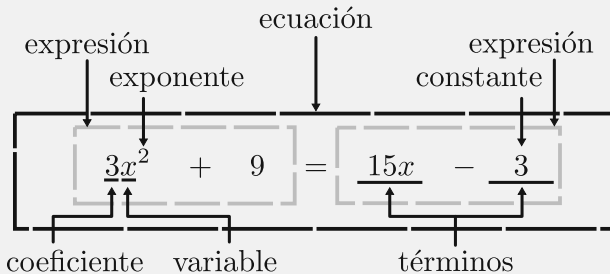
$$\log_b x = \frac{\log_k x}{\log_k b}$$

$$\ln x = \log_e x$$

$$\ln(e^x) = x \quad e^{\ln x} = x$$

$$\log_b 1 = 0$$

$$\log_b b = 1$$



$$\pi \approx 3.14159$$

$$e \approx 2.71828$$

$$\sqrt{2} \approx 1.41421$$

$$\varphi \approx 1.61803$$

$$\tau \approx 6.28318$$

$$ax^2 + bx + c = 0 \quad (a \neq 0)$$

$$\Delta = b^2 - 4ac$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Delta > 0 \quad 2 \text{ soluciones reales}$$

$$\Delta = 0 \quad 1 \text{ solución real}$$

$$\Delta < 0 \quad 0 \text{ soluciones reales}$$

$$(a + b)(a - b) = a^2 - b^2$$

$$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$$

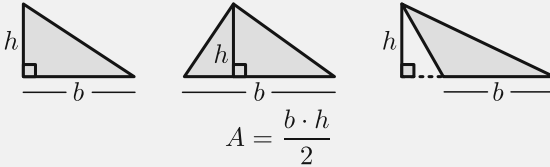
$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

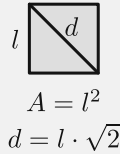
FIGURAS

@revecklife_

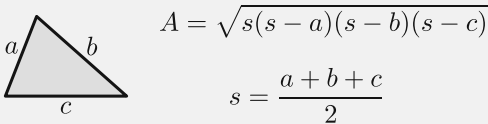
Triángulo



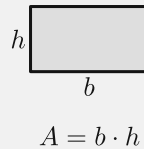
Cuadrado



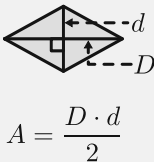
Fórmula de Herón



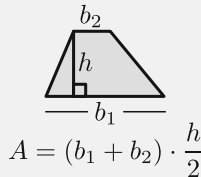
Rectángulo



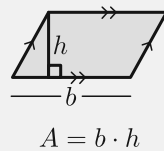
Rombo



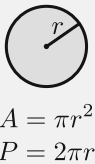
Trapecio



Paralelogramo

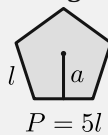


Círculo



Polígonos regulares

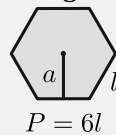
Pentágono



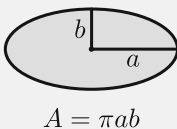
$$A = \frac{P \cdot a}{2}$$

— a: apotema —

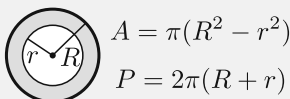
Hexágono



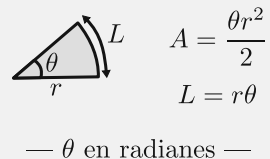
Elipse



Corona circular



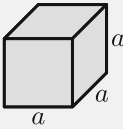
Sector circular



CUERPOS

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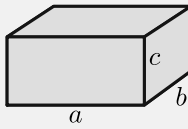
Cubo



$$A = 6a^2$$

$$V = a^3$$

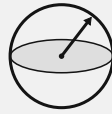
Ortoedro



$$A = 2(ab + ac + bc)$$

$$V = abc$$

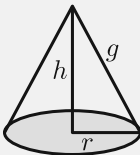
Esfera



$$A = 4\pi r^2$$

$$V = \frac{4}{3}\pi r^3$$

Cono

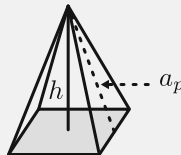


$$A_L = \pi r g$$

$$A_T = \pi r g + \pi r^2$$

$$V = \frac{1}{3}\pi r^2 h$$

Pirámide

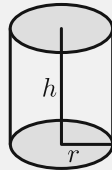


$$A_L = \frac{P_B \cdot a_p}{2}$$

$$A_T = A_L + A_B$$

$$V = \frac{1}{3}A_B \cdot h$$

Cilindro

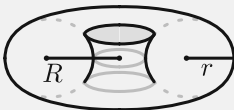


$$A_L = 2\pi r h$$

$$A_T = 2\pi r h + 2\pi r^2$$

$$V = \pi r^2 h$$

Toro (Dona)

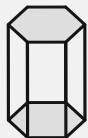


$$A = 4\pi^2 R r$$

$$V = 2\pi^2 R r^2$$

Prismas rectos

Triangular Cuadrangular Pentagonal Hexagonal



$$A_L = P_B \cdot h$$

$$A_T = A_L + 2A_B$$

$$V = A_B \cdot h$$

A_L = Área Lateral
 A_B = Área de la Base

A_T = Área Total
 g = Generatriz

P_B = Perímetro de la Base
 a_p = Apotema de la pirámide